

Supplementary Material for Texture Resonance Retrieval for Guitar-Effect Preset Selection

Anonymous Authors

S1 Canonical Evidence Sources and Metric Notes

This supplementary package is synchronized to the repository’s canonical experimental artifacts and the E9 derived evidence package used by the main manuscript. The primary objective evidence consists of the Protocol-A split audit, unified Protocol-A top-1 retrieval, Protocol-B exemplar-preserving projection, CLAP-covered shared-subset analysis, near-duplicate sensitivity, hard-split sweep, EPR sensitivity, EPR validity/repair audits, a dimensionality-matched control baseline, and cross-split stability checks on the same 204-query test split. We include the listening-study report as complementary subjective evidence. We do not use legacy comparison exports as formal evidence in this revision.

All objective metrics follow the same operational definitions used in the main paper. **Norm.L2** is RMSE over flattened numeric parameter leaves after plugin-range normalization; **Norm.Acc@0.1** measures the fraction of normalized numeric leaves within an absolute tolerance of 0.1; **Recall** measures retrieval over active numeric parameters; **Cosine** is cosine similarity in the flattened parameter space; and **SwitchF1** is the value-aware active-switch F1. The earlier key-presence **Module** Jaccard is retained only in JSON audits because it can reward complete-schema outputs without verifying switch values. Missing numeric keys are treated as zeros and non-numeric leaves are ignored. Accordingly, absolute magnitudes should be interpreted *within* each stated protocol rather than as perceptual distances.

S2 Protocol-A: Split Audit and Benchmark Definition

The external benchmark drop-in contains 1267 total records. A post-hoc audit of the original song-name-based held-out list (used in earlier drafts) revealed exact cross-split audio reuse, so the manuscript now anchors its main objective evidence to a stricter resolved-audio-grouped split. Table S1 summarizes both definitions.

Table S1: Protocol-A split audit summary. “Near-duplicate” counts come from the repository’s cheap fingerprint audit and therefore indicate a remaining risk surface rather than a formal perceptual-duplicate guarantee.

Split definition	N_{test}	N_{kb}	Shared audio paths	Near-duplicate pairs
Legacy song-name split	211	1056	16	48
Resolved-audio grouped split	204	1063	0	200

The stricter split groups records by resolved local audio path before sampling the held-out set, thereby eliminating exact cross-split audio reuse under the current audit. At the same time, the benchmark remains a pilot setting: the cheap fingerprint still reports many near-duplicate pairs, and the corpus remains guitar-only. The dataset JSON exposes **SongName**, **Style**, **Feature**, **Parameters**, and **Vectors**, but it does not include a separate auditable text-source field; accordingly, **Style** and **Feature** are the only explicit text-side provenance fields available in-repo. The stricter split contains 204 paired TRR rows but only 203 unique **query_name** labels because **Hard Rock Crunch** appears twice in the dataset; all pairing therefore uses **query_idx**.

S3 Unified Protocol-A/Protocol-B Main Comparison

Protocol-A compares top-1 executable retrieval on a shared 204-query split and a 1,063-record retrieval database. Protocol-B applies exemplar-preserving projection over the same retrieved neighborhoods. PaSST vectors were regenerated in the E9 derived artifact and pass the full 204-query/1,063-candidate coverage gate. PANNs remains a runtime blocker in the current Python 3.12 environment and is not claimed as a fair Protocol-B baseline.

Table S2: Unified Protocol-A/Protocol-B comparison. Protocol-A rows are top-1 retrieval; Protocol-B rows apply EPR-K5. Lower Norm.L2 is better; higher is better for the other metrics.

Method	Protocol	n	Coverage	Norm.L2 ↓	Norm.Acc@0.1 ↑	Recall ↑	Cosine ↑	SwitchF1 ↑	Prov.
Wav2Vec top-1	A	204	1.000	0.1997	0.7085	0.6228	0.8349	0.7692	yes
Wav2Vec EPR-K5	B	204	1.000	0.1838	0.7030	0.6194	0.8622	0.7886	EPR
FeatureNN top-1	A	204	1.000	0.2002	0.7055	0.6208	0.8435	0.7898	yes
FeatureNN EPR-K5	B	204	1.000	0.1878	0.6864	0.6135	0.8640	0.7970	EPR
CLAP top-1	A	173	0.848	0.1896	0.7204	0.6361	0.8538	0.8125	yes
CLAP EPR-K5	B	173	0.848	0.1642	0.7286	0.6447	0.8967	0.8553	EPR
PaSST top-1	A	204	1.000	0.1960	0.7063	0.6256	0.8463	0.7970	yes
PaSST EPR-K5	B	204	1.000	0.1808	0.6965	0.6170	0.8723	0.8123	EPR
TRR top-1	A	204	1.000	0.1454	0.7863	0.7045	0.9010	0.8630	yes
TRR EPR-K5	B	204	1.000	0.1334	0.7908	0.7146	0.9207	0.8824	EPR
MLP + RangeProjection	Direct	204	1.000	0.1603	0.7894	0.3468	0.5941	–	no

S4 Vector Coverage Gate

Table S3 records the vector coverage gate used before a method can enter Protocol-A/Protocol-B. A retrieval method must have all 204 query vectors and all 1,063 candidate vectors to be called full coverage. CLAP is reported with its actual coverage; PaSST passes after E9 regeneration; PANNs is excluded.

Table S3: Vector coverage gate for the unified evidence package.

Method	Query vectors	KB vectors	Ready	Main-table use	Boundary
TRR	204/204	1063/1063	yes	full coverage	proposed prior
Wav2Vec	204/204	1058/1063	usable	full-query audit	same-backbone baseline
FeatureNN	204/204	1058/1063	usable	full-query audit	dense reference
CLAP	173/204	829/1063	partial	coverage-limited	semantic prior
PaSST	204/204	1063/1063	yes	full coverage	modern audio encoder
PANNs	0/204	0/1063	no	excluded	runtime blocker

S5 TRR Mechanism Ablations

The mechanism audit includes one backbone-controlled comparison. TRR and Wav2Vec use the same Wav2Vec2 acoustic backbone, but differ in the aggregation operator. Wav2Vec uses first-order mean pooling, whereas TRR uses projected frame activations followed by Gram-matrix second-order statistics and L2 normalization. Table S4 isolates this comparison from the broader external-baseline matrix.

To further test the mechanism rather than only the final leaderboard, we recomputed controlled Wav2Vec2 layer features on the same 204-query archived split with a fixed random seed and compared same-backbone aggregation variants. This ablation is not a replacement for the user-verified archived E2 leaderboard because it uses the current local execution path and deterministic projection matrices. Its role is narrower: it checks whether the second-order design choices are directionally supported.

The mechanism result supports four bounded claims. First, Gram aggregation is not merely a Wav2Vec2-backbone effect: the same-layer Gram variants improve normalized distance over same-backbone mean

Table S4: Backbone-controlled diagnostic from archived Protocol-A. TRR and Wav2Vec share the Wav2Vec2 backbone but use different aggregation operators.

Method	Aggregation	Norm.L2 ↓	Acc@0.1 ↑	Recall ↑	Cosine ↑	Module ↑
TRR	Gram second-order statistics	0.1881	0.5766	0.5404	0.8415	0.8297
Wav2Vec	Mean pooling	0.2400	0.5015	0.4648	0.5642	0.7423

Table S5: Controlled TRR mechanism ablation on the archived split using the current local execution path. Lower is better for Norm.L2; higher is better for all other metrics.

Variant	Layers	Dim	Norm.L2 ↓	Acc@0.1 ↑	Recall ↑	Cosine ↑
Mean pooling	5	768	0.2582	0.6089	0.4623	0.5895
Mean pooling	4+5+6	768	0.2552	0.6061	0.4589	0.6111
Gram, no projection	5	589824	0.2501	0.6103	0.4635	0.6092
Gram, projected	4	4096	0.2528	0.6017	0.4585	0.6191
Gram, projected	5	4096	0.2457	0.6104	0.4654	0.6224
Gram, projected	6	4096	0.2529	0.5985	0.4485	0.6317
Gram, projected	4+5+6, $d = 16$	256	0.2501	0.6052	0.4449	0.6070
Gram, projected	4+5+6, $d = 32$	1024	0.2480	0.6022	0.4640	0.6398
Gram, projected	4+5+6, $d = 64$	4096	0.2546	0.5963	0.4487	0.6249
Gram, projected	4+5+6, $d = 128$	16384	0.2576	0.6054	0.4538	0.6121
Gram, no L2 norm	4+5+6, $d = 64$	4096	0.2849	0.5645	0.3944	0.5857

pooling. Second, the gain is strongest on a single mid-level layer (layer 5) rather than from multi-layer averaging: mean-pooling across layers 4–6 improves only marginally over mean-pooling layer 5 alone (0.2552 vs. 0.2582), and multi-layer Gram with $d = 64$ improves only marginally over multi-layer mean pooling (0.2546 vs. 0.2552). This means the reported TRR advantage is not experimentally isolated from the choice of layer 5 as the representative mid-level feature. Third, normalization is important; removing L2 normalization produces the weakest row in the table. Fourth, architectural choices remain meaningful: layer 5 and $d = 32/d = 64$ projected multi-layer variants trade off different metrics, so the paper should not claim that the reported TRR configuration is globally optimal. The generated artifact also records top-3 and top-5 oracle-best normalized L2 within each retrieval list, which we use only as a top- K sensitivity diagnostic rather than as the deployed top-1 metric. We therefore frame TRR as a useful texture-aware retrieval prior rather than as a fully optimized representation family.

S5.1 Dimensionality-Matched Control Baseline

A natural alternative explanation for the TRR advantage is that the Gram-matrix descriptor is higher-dimensional (4096-D) than the Wav2Vec mean-pooled baseline (768-D). To rule this out, we construct a dimensionality-matched control that projects the existing 768-D Wav2Vec vector to 4096-D via a fixed random Gaussian projection (seed 42, elements $\sim \mathcal{N}(0, 1/\sqrt{768})$) followed by L2 normalization. This control preserves the first-order information while matching the descriptor dimension of TRR. Table S6 reports the result on the original 204-query Protocol-A split.

Table S6: Dimensionality-matched control baseline. Wav2Vec.RP4096 projects the 768-D mean-pooled Wav2Vec vector to 4096-D. Lower Norm.L2 is better; higher is better for the other metrics.

Method	Dimension	Norm.L2 ↓	Acc@0.1 ↑	Recall ↑	Cosine ↑	SwitchF1 ↑
Wav2Vec	768-D	0.1997	0.7085	0.6228	0.8349	0.7692
Wav2Vec.RP4096	4096-D	0.1999	0.7074	0.6204	0.8338	0.7649
TRR	4096-D	0.1454	0.7863	0.7045	0.9010	0.8630

The 4096-D random projection achieves $\text{Norm.L2} = 0.1999$, statistically tied with the original 768-D baseline (0.1997) and far above TRR (0.1454). This confirms that the TRR advantage cannot be attributed to higher descriptor dimensionality alone.

S5.2 Cross-Split Stability

To verify that the main result is not an artifact of a single random split, we regenerate five independent splits under the same resolved-audio-grouped anti-leakage protocol with different random seeds. Table S7 reports top-1 Norm.L2 for TRR, Wav2Vec, and FeatureNN across splits.

Table S7: Cross-split stability on resolved-audio-grouped Protocol-A splits. Each split uses a different random seed while preserving the grouping constraint.

Split	n	TRR L2 ↓	Wav2Vec L2 ↓	FeatureNN L2 ↓	TRR rank
seed0 (original)	204	0.1454	0.1997	0.2002	1st
seed1	203	0.1543	0.1848	0.2088	1st
seed2	203	0.1501	0.1785	0.1927	1st
seed3	203	0.1495	0.1926	0.2026	1st
seed4	203	0.1507	0.1965	0.1919	1st
Mean	—	0.1500	0.1904	0.1992	—
Std	—	0.0032	0.0078	0.0068	—

TRR achieves the lowest Norm.L2 on all five splits, with a mean of 0.1500 ± 0.0032 . The Wav2Vec baseline varies more across splits (0.1904 ± 0.0078), reflecting its sensitivity to database density, yet it never approaches the TRR level.

S6 Experimental Figures and Qualitative Illustrations

This section collects the experiment figures used for qualitative inspection and auxiliary diagnostics. The workflow illustrations are not treated as result evidence; the quantitative claims remain anchored to the audited tables and data-backed diagnostic plots.

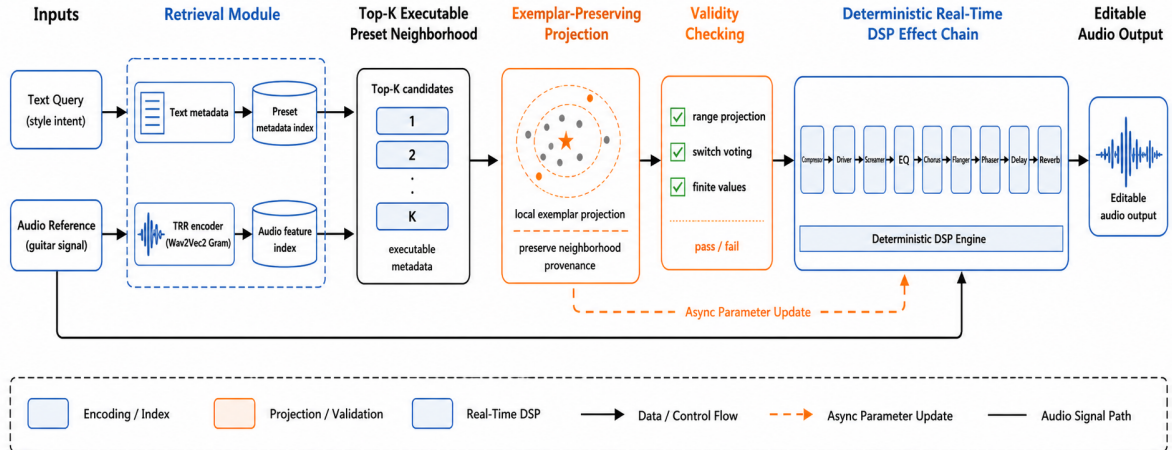


Figure S1: Retrieval-grounded guitar-effect preset-selection pipeline. This figure corresponds to the main-paper system overview and is repeated here for standalone supplementary readability.

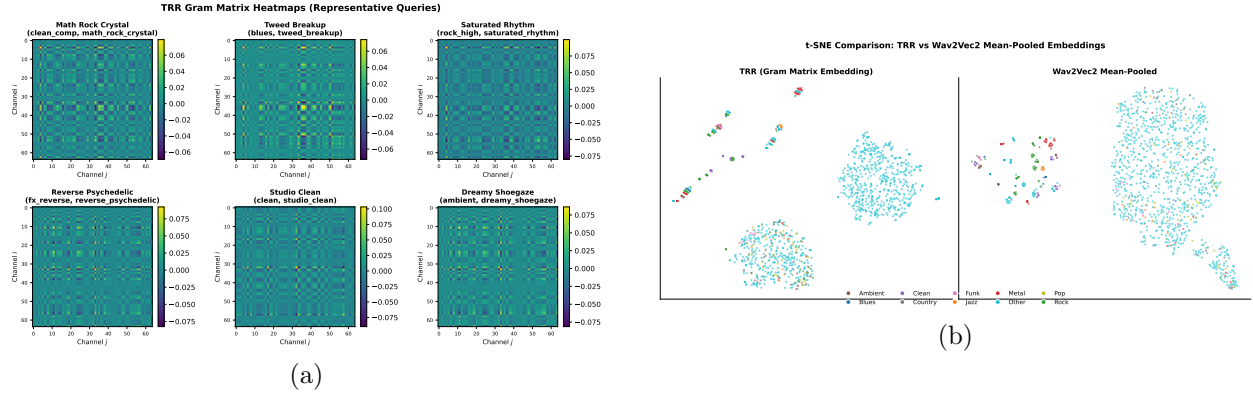


Figure S2: Representation diagnostics. (a) Example TRR Gram matrix. (b) t-SNE comparison between TRR and mean-pooled Wav2Vec2 embeddings. These plots are interpretive diagnostics and are not used as standalone proof of parameter transferability.

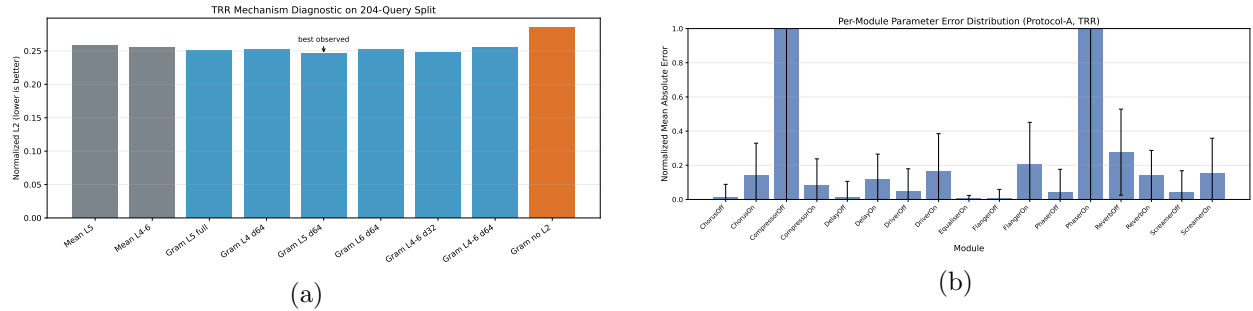


Figure S3: Mechanism and error diagnostics. (a) TRR layer-selection diagnostic. (b) Per-parameter error distribution diagnostic. These figures complement the mechanism ablation tables by showing where representation and parameter-level errors concentrate.

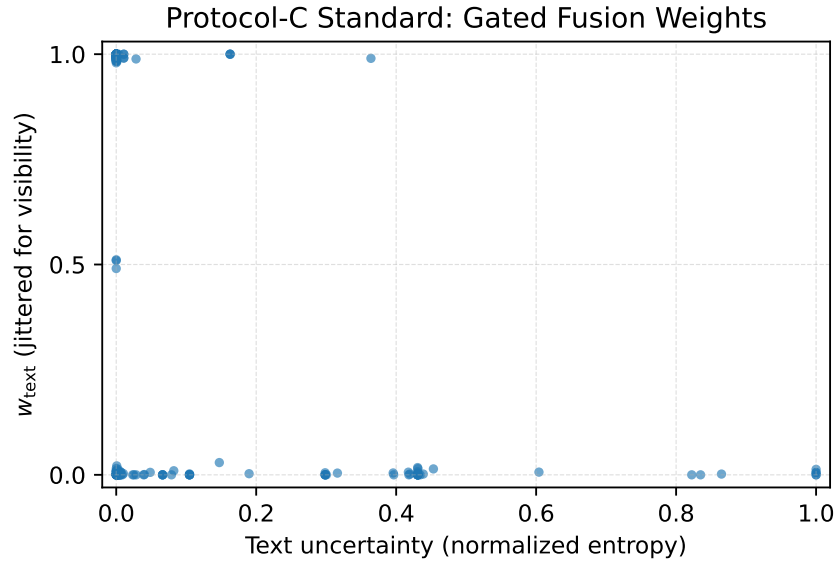


Figure S4: Fusion-weight distribution diagnostic for the degraded-input fusion experiment. The main manuscript treats adaptive fusion as a supplementary robustness-boundary diagnostic rather than evidence for the core TRR/EPR effectiveness claim.

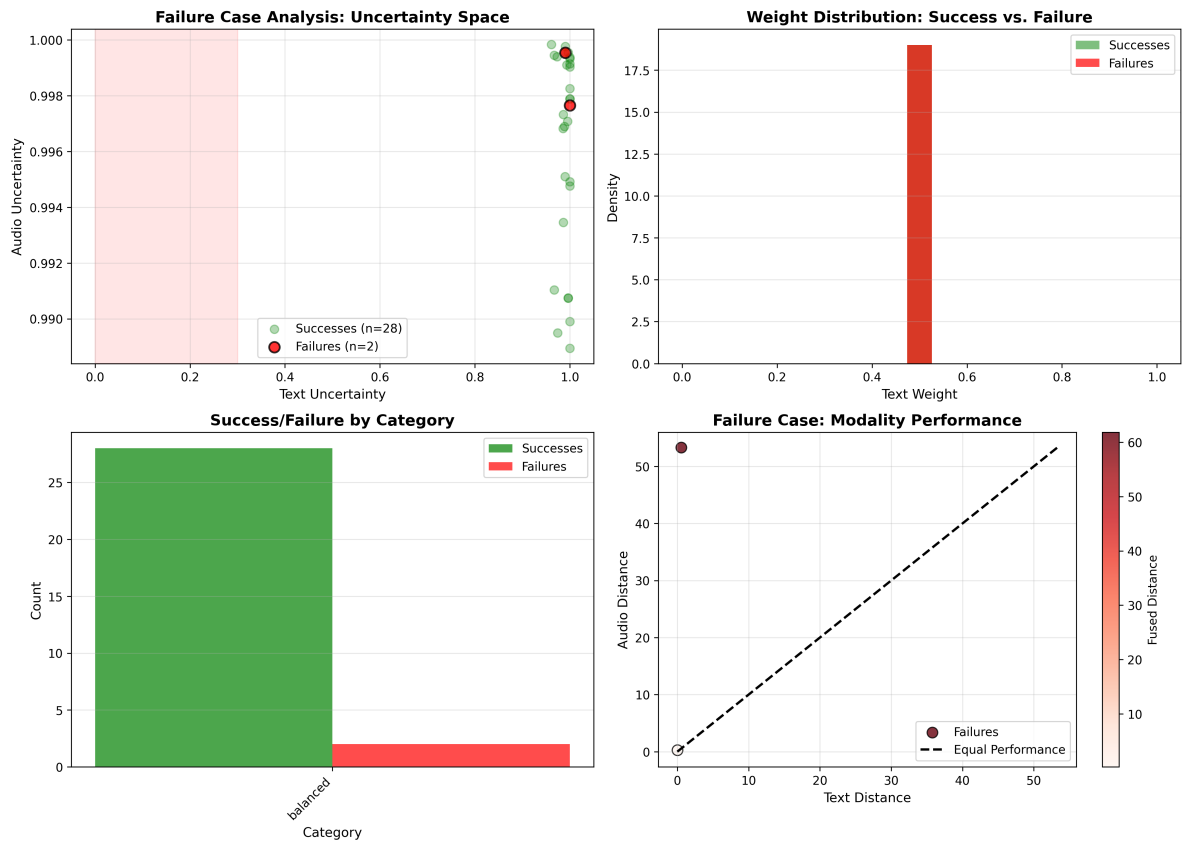


Figure S5: Fusion failure-case analysis. The plots inspect where adaptive fusion fails or succeeds in uncertainty space, category counts, and modality-distance comparisons. They delimit robustness behavior and do not support a dominant-fusion claim.

Adaptive Parameter Synthesis Agent Architecture

Figure S6: Representative adaptive-agent example used as a qualitative workflow illustration. It is included for interpretability and does not define the main evidence boundary.

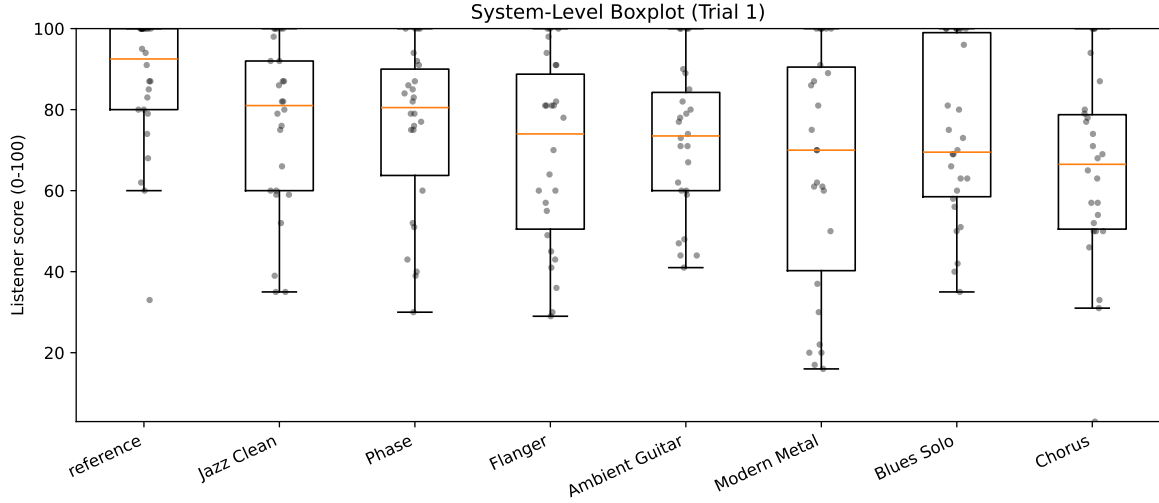


Figure S7: Listening study Trial 1: condition-level score distributions for style matching. The study is treated as perceptual context because it is not a standard MUSHRA test.

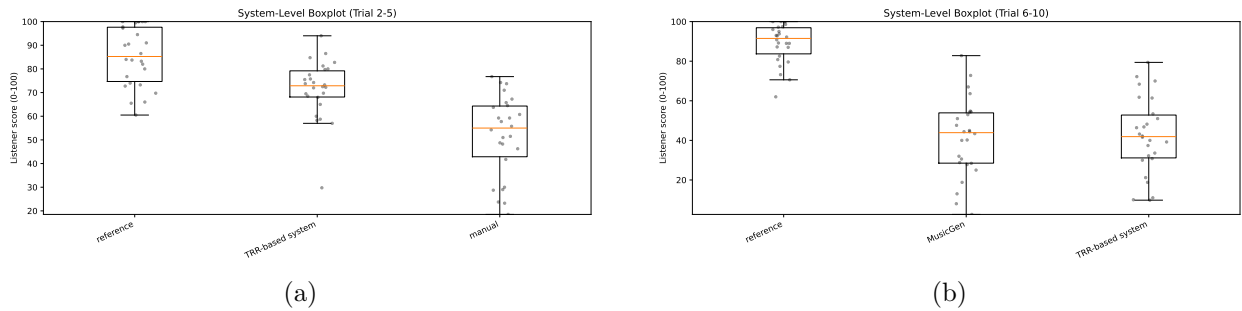


Figure S8: Listening study score distributions. (a) Trials 2–5 compare the TRR-based system with a manual baseline. (b) Trials 6–10 compare the TRR-based system with MusicGen. These plots are supporting perceptual diagnostics, not the primary Protocol-A/Protocol-B validation.

S7 Resolved-Audio-Grouped Protocol-A Audit Context

Before the archived baseline matrix was added, the revision used a resolved-audio-grouped Protocol-A table to avoid exact cross-split audio reuse. That table remains useful as audit context because it documents the anti-leakage split behavior and the earlier CLAP-overlap comparison. It should not be mixed numerically with the archived table because the reported L2 scale and baseline set differ.

Table S8: Resolved-audio-grouped Protocol-A audit table retained for split transparency.

Method	n	Param. Dist.	Acc@0.1	Recall	Cosine	Module
TRR (Ours)	204	8.0467	0.5169	0.4595	0.8247	0.8051
Text-RAG	204	24.1705	0.4717	0.3967	0.4943	0.7456
Wav2Vec-RAG	204	23.8197	0.4251	0.3505	0.5292	0.7043
FeatureNN-RAG	204	19.2613	0.4198	0.3600	0.5479	0.7134
CLAP	201	22.3102	0.4492	0.3971	0.5725	0.7240

S8 Protocol-A: Per-Query Diagnostics

To make the main benchmark less dependent on mean-only reporting, we derive per-query diagnostics from the canonical Protocol-A CSV for the resolved-audio-grouped split. The source artifact is the derived Protocol-A analysis file pair in Markdown and JSON form.

These diagnostics are paired by `query_idx`. The TRR rows contain 204 paired queries but only 203 unique `query_name` labels because one label (**Hard Rock Crunch**) is reused in the CSV; we therefore treat `query_idx` as the authoritative pairing key.

Table S9: Per-query win/loss/tie summary for TRR under Protocol-A. The interquartile range collapses to zero in many rows because several queries retrieve the same preset under both methods.

Baseline	Metric	Wins	Losses	Ties	Mean Δ
CLAP	Param. Dist.	130	54	17	14.1460
CLAP	Cosine	124	60	17	0.2503
CLAP	Module	79	39	83	0.0781
FeatureNN-RAG	Param. Dist.	130	49	25	11.2146
FeatureNN-RAG	Cosine	122	57	25	0.2769
FeatureNN-RAG	Module	86	40	78	0.0917
Text-RAG	Param. Dist.	117	69	18	16.1238
Text-RAG	Cosine	117	69	18	0.3304
Text-RAG	Module	73	56	75	0.0595
Wav2Vec-RAG	Param. Dist.	125	40	39	15.7730
Wav2Vec-RAG	Cosine	121	44	39	0.2956
Wav2Vec-RAG	Module	84	43	77	0.1008

Table S10: Representative Protocol-A L2 cases from the derived per-query analysis. Positive Δ indicates TRR is better.

Type	Baseline	Query	TRR retrieved	Baseline retrieved	Δ L2
Largest gain	CLAP	Modular Hybrid Drift	Prepared Guitar Static	Ethereal Wave	+71.6821
Largest gain	FeatureNN-RAG	Sub-Bass Drone	Hauntology Library	Dark Ambient Drone	+72.1369
				Prism	
Largest gain	Text-RAG	Sub-Bass Drone	Hauntology Library	Studio Clean	+72.2040
Largest gain	Wav2Vec-RAG	3-Chord Punk	Distorted Cassette	Britpop Jangle	+76.2671
Largest failure	CLAP	Talk Box Funk	Schlager Pop	Drum and Bass Synth	-57.3198
Largest failure	FeatureNN-RAG	Rave Hardcore	Dark Ambient	Melodic Death Metal	-56.3859
Largest failure	Text-RAG	Talk Box Funk	Schlager Pop	Studio Clean	-57.3623
Largest failure	Wav2Vec-RAG	Phased Synth	London Jazz	Nightcore Speed	-51.6878

S9 archived Protocol-C: Real-Degradation Robustness

archived Protocol-C evaluates realistic audio degradations by corrupting raw audio first and then re-encoding the degraded signal into TRR embeddings. It reuses the archived Protocol-A 204-query test split, uses **Style+Feature** as the text branch, and computes ranking relevance from parameter-space oracle neighbors rather than exact song-name matching.

Table S11: archived Protocol-C overall summary on the shared 204-query split. Lower is better for L2; higher is better for all other metrics.

Method	α	L2 $\downarrow$	Acc@0.1 $\uparrow$	MRR $\uparrow$	NDCG@5 $\uparrow$
Fixed	0.3000	29.2322	0.4846	0.4095	0.3723
Fixed	0.5000	20.5551	0.6085	0.5607	0.5528
Fixed	0.7000	20.1868	0.6104	0.5652	0.5551
Adaptive fusion	0.5980	20.1491	0.6121	0.5654	0.5566

Table S12: archived Protocol-C per-condition winners across the main multimodal comparison set.

Condition	Best L2	Best Acc@0.1	Best MRR	Best NDCG@5
AWGN 20 dB	fixed 0.70	adaptive	adaptive	adaptive
AWGN 10 dB	fixed 0.70	adaptive	adaptive / fixed 0.50	fixed 0.50
AWGN 5 dB	fixed 0.70	fixed 0.70	fixed 0.70	fixed 0.70
MP3 128k	fixed 0.70	fixed 0.50	fixed 0.50	fixed 0.50
MP3 64k	fixed 0.70	fixed 0.50	fixed 0.50	fixed 0.50
MP3 32k	adaptive	fixed 0.50	adaptive / fixed 0.50	fixed 0.50
Reverb 0.6	adaptive	fixed 0.50	adaptive / fixed 0.50	adaptive
Reverb 1.0	adaptive	fixed 0.50	adaptive / fixed 0.50	fixed 0.50
Truncate 50%	adaptive	adaptive	adaptive / fixed 0.70	fixed 0.70
Truncate 30%	adaptive / fixed 0.70	adaptive	adaptive / fixed 0.70	adaptive / fixed 0.70

The per-condition view shows that adaptive fusion is not a trivial uniform winner; fixed 0.50 and fixed 0.70 remain competitive on several conditions. The stronger conclusion is that audio-heavy fusion is fragile, whereas adaptive fusion consistently moves the system toward the stable mid-to-high text-weight region while preserving multimodal behavior.

Local boundary audit. We additionally ran a Protocol-C boundary script with explicit audio-only ($\alpha = 0.0$), pure-text ($\alpha = 1.0$), and parameter-oracle upper-bound rows. Because this run follows a separate local

execution path from the archived summary artifact, we report it as an audit boundary rather than as a replacement for Table S11. The result in Table S13 confirms the main qualitative interpretation: audio-only retrieval is weak under realistic degradation, mid/high text weights are stable, and the pure-text boundary remains a strong practical boundary. The oracle row is not a method; it selects the nearest KB item using ground-truth parameters and only quantifies remaining headroom. Thus, adaptive fusion should be read as a multimodal operating strategy near the stable region, not as proof of superiority over text-only retrieval.

Table S13: Local-audit Protocol-C boundary audit with audio-only and pure-text rows. This table is an execution-path audit artifact, not a replacement for the canonical archived server summary.

Method	α	L2 ↓	Acc@0.1 ↑	MRR ↑	NDCG@5 ↑
Audio-only	0.0000	102.5577	0.4962	0.1184	0.0731
Fixed	0.3000	81.0390	0.5980	0.3916	0.3588
Fixed	0.5000	66.9431	0.6791	0.5633	0.5463
Fixed	0.7000	63.9030	0.6832	0.5730	0.5528
Adaptive fusion	0.5980	65.4614	0.6837	0.5719	0.5529
Pure text	1.0000	61.5079	0.6872	0.5847	0.5586
Parameter oracle upper bound	oracle	1.3284	0.9548	1.0000	1.0000

S10 Listening Study Statistics

The listening study uses a multiple-stimulus protocol with a hidden reference and no explicit low-quality anchor. The raw logs label our system as **TRR-based system**; throughout the manuscript and this supplement we refer to the same condition as the *TRR-based system*. Across all trials, the study contains 26 unique participants and 910 ratings.

The current logs preserve repeated-measures structure and participant IDs, but they do not preserve device metadata, participant expertise labels, or the provenance and time budget of the manual baseline. We therefore interpret Trials 2–5 as perceptual context rather than as a fairness-critical human-vs-system optimization benchmark.

Table S14: Study-level listening-test summary.

Trial block	Rows	Participants	Stimuli	Raw mean	Raw median / std
Trial 1	208	26	8	72.92	77.50 / 22.72
Trials 2–5	312	26	3	69.53	73.50 / 23.53
Trials 6–10	390	26	3	57.54	59.00 / 32.24

Table S15: Repeated-measures nonparametric tests for the listening study.

Block	Comparison	n	Mean(A-B)	Median(A-B)	rbc	p (Holm)
Trials 2–5	Friedman omnibus ($Q = 41.6154$, $p_{\text{perm}} = 5 \times 10^{-5}$)	N/A	N/A	N/A	N/A	N/A
Trials 2–5	reference vs TRR-based system	26	13.78	8.38	0.860	0.00015
Trials 2–5	reference vs manual	26	33.61	26.00	0.983	0.00015
Trials 2–5	TRR-based system vs manual	26	19.83	17.25	1.000	0.00015
Trials 6–10	Friedman omnibus ($Q = 36.2308$, $p_{\text{perm}} = 5 \times 10^{-5}$)	N/A	N/A	N/A	N/A	N/A
Trials 6–10	reference vs MusicGen	26	47.72	50.90	1.000	0.00015
Trials 6–10	reference vs TRR-based system	26	46.71	51.30	0.994	0.00015
Trials 6–10	MusicGen vs TRR-based system	24	-1.10	0.80	0.067	0.7797

Table S16: Trial 1 style-matching stimulus labels used in the listening study.

Label	Interpretation
Jazz Clean	clean jazz tone target
Blues Solo	blues lead tone target
Phase	phaser/phase-effect target
Ambient Guitar	ambient guitar texture target
Flanger	flanger-effect target
Modern Metal	modern high-gain tone target
Chorus	chorus-effect target

S11 Near-Duplicate Sensitivity Analysis

To assess the impact of near-duplicate presets on retrieval metrics, we progressively remove candidates whose normalized parameter RMSE to any held-out query is at most a threshold and re-evaluate the stored-vector retrieval methods under the same unified evaluator as the main table. Threshold 0.000 is explicitly defined as the no-filter canonical row; therefore the TRR, Wav2Vec, FeatureNN, CLAP, and PaSST values in that row match the corresponding Protocol-A rows in the main table.

Table S17: Near-duplicate sensitivity under the unified Protocol-A evaluator. Lower Norm.L2 is better.

Threshold	KB size	TRR	Wav2Vec	FeatureNN	CLAP	PaSST
0.000	1063	0.1454	0.1997	0.2002	0.1896	0.1960
0.005	599	0.1933	0.2281	0.2312	0.2147	0.2293
0.010	342	0.1968	0.2417	0.2503	0.2422	0.2542
0.020	151	0.2510	0.2571	0.2836	0.2687	0.2758
0.050	117	0.2932	0.3048	0.3104	0.3113	0.3176

S12 CLAP-Covered Shared Subset

CLAP coverage is partial in the local JSON artifact, so we recompute a matched comparison on the intersection where Wav2Vec, FeatureNN, CLAP, PaSST, and TRR all have vectors for both queries and candidate presets. The resulting shared subset contains 173 queries and 829 candidate presets.

Table S18: CLAP-covered shared query and knowledge-base subset. Lower Norm.L2 is better; higher PNR@5 is better.

Method	n	Top-1 Norm.L2	PNR@5	EPR-K5 Norm.L2
Wav2Vec	173	0.1939	0.6705	0.1741
FeatureNN	173	0.1962	0.6763	0.1794
CLAP	173	0.1896	0.7803	0.1642
PaSST	173	0.1894	0.7283	0.1645
TRR	173	0.1487	0.8150	0.1328

S13 Parameter-Cluster Hard Split

The main text reports a compact hard-split table. This supplementary section records bootstrap 95% confidence intervals for the same sweep. Starting from the 204-query Protocol-A list, we cluster records whose normalized parameter RMSE is at most τ_{cluster} and remove every knowledge-base candidate in a selected query cluster. This is intentionally severe: it tests duplicate-family sensitivity under a small candidate pool and should not replace the unified main table.

Table S19: Parameter-cluster hard-split sweep with bootstrap 95% confidence intervals for normalized L2. CLAP has partial query coverage.

τ_{cluster}	Method	n	KB	clusters	Norm.L2	95% CI
0.01	TRR	50	278	130	0.2065	[0.1730, 0.2439]
0.01	CLAP	37	278	130	0.1911	[0.1532, 0.2283]
0.01	PaSST	50	278	130	0.2413	[0.2047, 0.2782]
0.01	Wav2Vec	50	278	130	0.2203	[0.1824, 0.2553]
0.01	FeatureNN	50	278	130	0.2188	[0.1839, 0.2532]
0.02	TRR	46	142	86	0.2424	[0.2131, 0.2732]
0.02	CLAP	35	142	86	0.2065	[0.1665, 0.2463]
0.02	PaSST	46	142	86	0.2540	[0.2149, 0.2888]
0.02	Wav2Vec	46	142	86	0.2595	[0.2291, 0.2876]
0.02	FeatureNN	46	142	86	0.2634	[0.2274, 0.2980]
0.05	TRR	38	117	67	0.2668	[0.2350, 0.3012]
0.05	CLAP	29	117	67	0.2782	[0.2463, 0.3092]
0.05	PaSST	38	117	67	0.3162	[0.2845, 0.3476]
0.05	Wav2Vec	38	117	67	0.2822	[0.2510, 0.3138]
0.05	FeatureNN	38	117	67	0.2771	[0.2439, 0.3116]

The hard split changes the interpretation of the retrieval results. Among full-coverage rows, TRR remains the best normalized-distance method in all three sweeps. CLAP is stronger on its covered subset for two thresholds, which is why the main text treats the hard split as a diagnostic stress test rather than as a dominance proof.

S14 Uncached Latency Audit

The cached latency table in the main text measures retrieval scoring after embeddings are already available. A separate audit of the reviewed artifact set found no verified live path that jointly performs audio preprocessing, Wav2Vec2 forward inference, TRR re-encoding, search, and validation for newly uploaded reference audio. Therefore, the paper uses the cached latency table only as a retrieval-scoring diagnostic and does not claim end-to-end live reference-audio latency.

S15 Additional Baseline Status

The E9 derived artifact closes the PaSST retrieval-baseline gap for the shared Protocol-A/Protocol-B benchmark. PANNs remains excluded because the local inference dependency chain is blocked under Python 3.12. We also include a diagnostic direct-regression baseline using mean-pooled Wav2Vec2 features and a two-layer MLP trained only on the knowledge base. This regressor is useful as a boundary condition, but it is not

Table S20: Uncached latency evidence status for newly supplied reference audio.

Required component	Evidence status
Audio preprocessing	Not verified in artifact
Wav2Vec2 forward pass	Not verified in artifact
TRR re-encoding	Not verified in artifact
Nearest-neighbor search	Cached-path timing available
Parameter validation	Not verified in uncached path

an executable-preset retriever: it predicts dense continuous parameter leaves and still requires constraint projection before deployment. Learned reranking remains future work.

Table S21: Additional baseline status under the E9 unified evidence package.

Method	n	Evidence status	Use in paper	Boundary
PaSST	204	E9 complete	Main unified table	Full-coverage modern audio encoder
PANNs	0	Runtime blocked	Excluded	No fair Protocol-B claim
MLP-Regressor	204	Diagnostic complete	Boundary baseline	Dense non-executable output

The range-projected MLP row obtains normalized L2 0.1603, Norm.Acc@0.1 0.7894, Recall 0.3468, and Cosine 0.5941 on 204 test queries under the unified exporter. This mixed profile is informative: the low recall and cosine score indicate that direct regression does not recover the same active parameter structure as retrieval. The legacy key-presence Module Jaccard is not used as a main conclusion because it can become artificially high for dense schema-filling methods. Under the current Protocol-B diagnostic, the range-projected MLP is weaker than TRR top-1 and EPR-K5 on Norm.L2; its role is therefore boundary evidence for non-executable regression, not a stronger numeric method.

S16 Protocol-B Executable-Neighborhood Diagnostics

Protocol-B evaluates whether retrieval produces a parameter-transferable executable neighborhood rather than only a single nearest preset. **PNR@K** is parameter-neighborhood recall at threshold 0.10 over the retrieved set in the main tables. EPR denotes exemplar-preserving projection: a softmax-weighted blend over the retrieved executable presets followed by range and switch projection, with provenance reported as effective exemplars and maximum exemplar weight. Module-aware reranking is included only as a diagnostic negative result because it worsens Norm.L2 and degrades recall, cosine, and legacy module alignment in the current run.

Table S22: Protocol-B executable-neighborhood diagnostics on the Protocol-A query list under the unified execution path. For pure TRR rows, Norm.L2 and Norm.Acc@0.1 are top-1 retrieval metrics; TRR top-K reports neighborhood recall. Provenance is reported as effective exemplars / max weight.

Method	K	PNR@K	Norm.L2	Norm.Acc@0.1	SwitchF1	Provenance
TRR top-1	1	0.4608	0.1454	0.7863	0.8630	yes
TRR top-K	5	0.7500	—	—	—	top-K list
EPR-K3	3	—	0.1356	0.7872	0.8734	2.6857 / 0.4664
EPR-K5	5	—	0.1334	0.7908	0.8824	4.2129 / 0.3438
MLP + RangeProjection	—	—	0.1603	0.7894	—	none
Module-aware rerank	5	—	0.1479	0.7870	—	diagnostic negative

The primary hybrid row is EPR-K5, not the range-projected MLP. EPR-K5 gives the best normalized distance in this diagnostic table while retaining neighborhood provenance (every output value is bounded by the ranges and votes of the retrieved top-5 exemplars). The range-projected MLP remains useful as a direct-regression boundary row, but it has no retrieved-neighborhood provenance.

Table S23: TRR EPR sensitivity. K=5, $\tau = 0.05$, softmax weighting is the primary manuscript row.

Weighting	K	τ	Norm.L2	Recall	Cos	SwitchF1	N_{eff}	max w
softmax	1	0.05	0.1454	0.7045	0.9010	0.8630	1.0000	1.0000
softmax	3	0.05	0.1356	0.7060	0.9177	0.8734	2.6857	0.4664
softmax	5	0.05	0.1334	0.7146	0.9207	0.8824	4.2129	0.3438
softmax	10	0.05	0.1307	0.7156	0.9258	0.8924	7.5900	0.2400
uniform	5	–	0.1379	0.7036	0.9157	0.8744	5.0000	0.2000
uniform	10	–	0.1324	0.7060	0.9274	0.8984	10.0000	0.1000

Table S24: Protocol-B EPR-K5 validity and repair audit. Validity is measured after range and switch projection.

Method	n	Validity	Repair	Range repair	Switch repair
Wav2Vec EPR-K5	204	1.0000	0.8627	0.5343	0.8627
FeatureNN EPR-K5	204	1.0000	0.9363	0.5588	0.9363
CLAP EPR-K5	173	1.0000	0.9306	0.5376	0.9306
PaSST EPR-K5	204	1.0000	0.9510	0.5931	0.9510
TRR EPR-K5	204	1.0000	0.7843	0.4363	0.7843

S17 Trials 2–5: Detailed Per-Trial Results

The main text reports only the aggregate Trials 2–5 results. Table S25 provides per-trial stimulus labels and per-condition ratings for completeness.

Table S25: Trials 2–5 per-trial breakdown. Ratings are raw means $\pm$ standard deviation across 26 participants.

Trial	Stimulus	Reference	TRR-based	Manual
Trial 2	Jazz Clean	91.19 $\pm$ 10.52	75.12 $\pm$ 13.99	53.88 $\pm$ 22.90
Trial 3	Blues Solo	85.12 $\pm$ 21.50	70.04 $\pm$ 17.28	58.54 $\pm$ 19.01
Trial 4	Phase	83.58 $\pm$ 18.74	66.88 $\pm$ 18.94	49.04 $\pm$ 19.35
Trial 5	Ambient Guitar	81.42 $\pm$ 19.66	74.15 $\pm$ 15.57	45.42 $\pm$ 25.44